

KALEY BRAUER

NSF Astrophysics Postdoctoral Fellow
Institute for Theory and Computation, Harvard University
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EDUCATION

Ph.D. in Physics	2023
Massachusetts Institute of Technology Astrophysics Division GPA: 5.0/5.0	
Sc.B. in Physics	2017
Brown University University honors, departmental honors, GPA: 4.0/4.0	

ACADEMIC POSITIONS

AI Alignment Research Fellow	2026 – present
Cambridge Boston Alignment Initiative; Mentor: Samuel Marks at Anthropic	
NSF Astrophysics Postdoctoral Fellow	2023 – present
Harvard University	
US Department of Energy Graduate Fellow; Graduate Researcher	2017 – 2023
Massachusetts Institute of Technology	
Visiting Research Fellow	Fall 2021
Lawrence Berkeley National Laboratory	

HONORS & AWARDS

Research Corporation for Science Advancement Fellow	2025
NSF Astronomy & Astrophysics Postdoctoral Fellowship (\$330K; ~8 awarded annually)	2023
Spot Awards, MIT, <i>in recognition of community service to School of Science</i>	2020 & 2022
DOE Computational Science Graduate Fellowship (~\$450K; ~25 awarded annually)	2018
NSF Graduate Research Fellowship (~ \$150,000)	2018
MIT Whiteman Fellowship (~\$100K)	2017
R. Bruce Lindsay Award, Brown University, <i>given to senior for excellence in physics</i>	2017
Eva A. Mooar Prize, Brown University, <i>given to woman for academic excellence</i>	2017
Sigma Xi Research Honor Society	2017
Karen T. Romer Undergraduate Teaching and Research Award, Brown University	2015

PUBLICATIONS

Refereed

Journal Articles

17. Emami, R., Trussler, J. A. A., Hsiao, T. Y.-Y., **Brauer, K.**, Hernquist, L., Smith, R., Finkbeiner, D., Sun, F., Davies, R., Steiner, J. F., Vogelsberger, M., Looser, T., Tremblay, G., Bugiani, L. (2026). Unraveling Chemical Enrichment in Extreme Emission-Line Galaxies: A Multi-Element

Bayesian View of Bursty Star Formation and Galaxy Evolution in DESI. *submitted to Journal of Cosmology and Astroparticle Physics*. arXiv:2604.20060

16. Cannon, R. E., Rozek, A., **Brauer, K.**, Busch, M. W., Snodgrass, C., Benner, L. A. M., Brozovic, M., Howell, E., Nolan, M. C., Rabus, M., Sajadian, S., Springmann, A., Taylor, P., Zambrano-Marin, L. F. (2026). The shape and spin state of (275677) 2000 RS11 from ground-based radar and optical observations. *in press at the Monthly Notices of the Royal Astronomical Society*. arXiv:2603.16703
15. Andersson, E. P., Rey, M. P., Yates, R. M., Read, J. I., Agertz, O., Ji, A. P., Mead, J., **Brauer, K.**, Mac Low, M. (2026). EDGE-INFERN0: The impact of chemical enrichment assumptions on the individual stars of a simulated faint dwarf galaxy. *in press at The Astrophysical Journal*. arXiv:2511.05695
14. Hazlett, R., Mead, J., Visbal, E., Bryan, G., Mac Low, M., Kulkarni, M., Andersson, E. P., **Brauer, K.**, Wise, J. (2025). From Primordial Stars to Early Galaxies: A Semi-Analytic Model Calibrated with AEOS and Renaissance. *submitted to The Astrophysical Journal*. arXiv:2510.11629
13. Mead, J., **Brauer, K.**, Bryan, G. L., Mac Low, M. M., Ji, A. P., Wise, J. H., Andersson, E. P., Frebel, A., Emerick, A., Cote, B. (2025). AEOS is Mixing it Up: The (In)homogeneity of Metal Mixing Following Population III Star Formation. *in press at The Astrophysical Journal*. arXiv:2509.13580
12. **Brauer, K.**, Mead, J., Ji, A. P., Wise, J. H., Bryan, G. L., Mac Low, M. M., Emerick, A., Andersson, E. P., Frebel, A., Cote, B. (2025). AEOS: The Impact of Pop III Initial Mass Function and Star-by-Star Models in Galaxy Simulations. *The Astrophysical Journal*, 993, 2.
11. Mead, J., **Brauer, K.**, Bryan, G. L., Mac Low, M. M., Ji, A. P., Wise, J. H., Emerick, A., Andersson, E. P., Frebel, A., Cote, B. (2025). AEOS: Transport of metals from minihalos following Population III stellar feedback. *The Astrophysical Journal*, 980, 62.
10. **Brauer, K.**, Emerick, A., Mead, J., Ji, A. P., Wise, J. H., Bryan, G. L., Mac Low, M. M., Cote, B., Andersson, E. P., Frebel, A. (2025). AEOS: Star-by-Star Cosmological Simulations of Early Chemical Enrichment and Galaxy Formation. *The Astrophysical Journal*, 980, 41.
9. Chiti, A., Mardini, M. K., Limberg, G., Frebel, A., Ji, A. P., Reggiani, H., Ferguson, P., Andales, H. D., **Brauer, K.**, Li, T. S., Simon, J. D. (2024). Signatures of Extragalactic First Stars in the Large Magellanic Cloud. *Nature Astronomy*, 8, 637647.
8. **Brauer, K.**, Andales, H., Ji, A. P., Mardini, M., Frebel, A., Gomez, F. A., & O'Shea, B. W. (2022). Possibilities and Limitations of Kinematically Identifying Stars from Accreted Ultra-Faint Dwarf Galaxies. *The Astrophysical Journal*, 937, 14.
7. Ji, A. P., Naidu, R., **Brauer, K.**, Ting, Y., & Simon, J. (2022). Chemical Abundances of the Typhon Stellar Stream. *Monthly Notices of the Royal Astronomical Society*, 519, 4467-4478.
6. Mardini, M., Frebel, A., Chiti, A., Meiron, Y., **Brauer, K.**, Ou, X. (2022). Characterization of the Metal Weak Thick Disk of the Milky Way. *The Astrophysical Journal*, 936, 78.
5. **Brauer, K.**, Ji, A. P., Drout, M., Frebel, A. (2021). Collapsar R-Process Yields Can Reproduce [Eu/Fe] Abundance Scatter in Metal-Poor Stars. *The Astrophysical Journal*, 915, 81.
4. Gull, M., Frebel, A., Hinojosa, K., Roederer, I. U., Ji, A. P., **Brauer, K.** (2021). R-process-rich stellar streams in the Milky Way. *The Astrophysical Journal*, 912, 52.

3. **Brauer, K.**, Ji, A. P., Frebel, A., Dooley, G. A., Gomez, F. A., & O’Shea, B. W. (2019). The Origin of r-process Enhanced Metal-Poor Halo Stars In Now-Destroyed Ultra-Faint Dwarf Galaxies. *The Astrophysical Journal*, 871, 247.
2. **Brauer, K.**, Vrtillek, S. D., Peris, C., & McCollough, M. (2018). Phase-resolved spectroscopy of the low-mass X-ray binary V801 Ara. *Monthly Notices of the Royal Astronomical Society*, 478, 4894-4904.

Refereed Workshop Papers

1. **Brauer, K.**, Dash, A. P., Vyas, M. J., Salim, A. (2025). From Simulations to Surveys: Domain Adaptation for Galaxy Observations. *Machine Learning and the Physical Sciences (ML4PS) Workshop at NeurIPS 2025*. arXiv:2511.18590

Non-Refereed

2. **Brauer, K.** (2021). “I’ll Finish it This Week” And Other Lies. *arXiv April Fools Paper*. arXiv:2103.16574
1. **Brauer, K.**, Ji, A., Hattori, K., Escobar, S., & Frebel, A. (2019). Kinematics of highly r-process-enhanced halo stars: Evidence for origins in now-destroyed ultra-faint dwarf galaxies. *Proceedings of the International Astronomical Union*, 14(S353), 71-74.

PRESENTATIONS

INVITED

AstroAI Workshop Talk at Harvard	Cambridge, MA, USA; Jun 2026
Astrophysics Theory Seminar Talk at UC San Diego	San Diego, CA, USA; Feb 2026
Talk at From Galaxies to Planets: An Elemental Journey	Ringberg Castle, Germany; Feb 2026
Physics Colloquium Talk at Rensselaer Polytechnic Institute (RPI)	Troy, NY, USA; Oct 2025
Astronomy & Data Science Talk at Yale University	New Haven, CT, USA; May 2025
Astronomy Seminar Talk at Columbia University	New York, NY, USA; Mar 2025
Fundamental Physics Seminar Talk at Brown University	Providence, RI, USA; Mar 2025
Astronomy Seminar Talk at Lund University	Lund, Sweden; Mar 2025
Astronomy Seminar Talk at American Museum of Natural History	New York, NY, USA; Oct 2024
Nuclear Astrophysics Seminar Talk at IReNA	East Lansing, MI, USA; Sep 2024
Colloquium Summer Talk at Harvard-Smithsonian CfA	Cambridge, MA, USA; July 2023
Talk at US Department of Energy CSGF Program Review	Washington, DC, USA; July 2022
High Performance Computing Talks at IHPCSS	Athens, Greece; June 2022
Colloquium Talk at University of Melbourne	Melbourne, Australia; Oct 2021
Seminar Talk at Computational Research in Boston and Beyond	Boston, USA; Oct 2021
Seminar Talk at Carnegie Observatories	Pasadena, CA, USA; Oct 2020

CONTRIBUTED

Poster at ICML 2026 Mech Interp Workshop	Seoul, South Korea; Jul 2026
Talk at NSF Postdoctoral Fellows Symposium 2026	Phoenix AZ, USA; Jan 2026

Poster at NeurIPS 2025 ML4PS Workshop	San Diego, CA, USA; Dec 2025
Talk at IAU Symposium 403 on Galactic Outskirts	Cordoba, Spain; Oct 2025
Poster at Tracing Cosmic Evolution with Clusters of Galaxies V	Sesto, Italy; July 2025
Talk at Galactic Frontiers II at Dartmouth	Hanover, CT, USA; Jun 2025
Talk at First Galaxies Meeting at Oxford	Oxford, UK; Apr 2025
Talk at NSF Postdoctoral Fellows Symposium 2025	Washington DC, USA; Jan 2025
Talk at IAU Symposium 395 on Stellar Populations	Paraty, Brazil; Nov 2024
Talk at Astrophysical Origins of Carbon	Tokyo, Japan; Sep 2024
Talk at First Stars VII at CCA	NYC, NY, USA; May 2024
Talk at Galaxies from Scratch 2024	Vienna, Austria; Feb 2024
Talk at 243rd Meeting of the American Astronomical Society	New Orleans, LA, USA; Jan 2024
Talk at NSF Postdoctoral Fellows Symposium 2024	New Orleans, LA, USA; Jan 2024
Talk at Flatiron CCA Galactic Frontiers	NYC, NY, USA; July 2023
Talk at IAU Symposium 377 on Early Disk Galaxy Formation	Kuala Lumpur, Malaysia; Feb 2023
Talk at 241st Meeting of the American Astronomical Society	Seattle, WA, USA; Jan 2023
Talk at JINA-CEE Frontiers in Nuclear Astrophysics Meeting	South Bend, IN, USA; May 2022
Talk at 2021 GALAH Science Meeting	Sydney, Australia; June 2021
Talk at <i>Linking the Galactic and Extragalactic</i>	Wollongong, Australia; Dec 2020
Talk at 235th Meeting of the American Astronomical Society	Honolulu, HI, USA; Jan 2020
Talk at IAU Symposium 353 on Galactic Dynamics	Shanghai, China; July 2019
Poster at JINA-CEE Frontiers in Nuclear Astrophysics Meeting	East Lansing, MI, USA; May 2019
Talk at JINA-CEE Frontiers in Nuclear Astrophysics Meeting	South Bend, IN, USA; May 2018
Poster at 229th Meeting of the American Astronomical Society	Grapevine, TX, USA; Jan 2017
Poster at 47th Meeting of the Division of Planetary Science	Washington, DC, USA; Nov 2015
Talk at SETI Institute's Summer Colloquium Series	Mountain View, CA, USA; Aug 2015

TECHNICAL SKILLS

Programming	Python, C/C++, MPI, OpenMP, GPU acceleration, Slurm, Git
Machine Learning	PyTorch, TensorFlow, HuggingFace, mechanistic interpretability

TEACHING

8.03 Vibrations and Waves Teaching Assistant	Spring 2023
<i>Physics Department, Massachusetts Institute of Technology</i>	
8.S30 Stellar Archaeology Teaching Assistant	Fall 2022
<i>Physics Department, Massachusetts Institute of Technology</i>	
High Performance Computing Summer School	Summer 2022
<i>International High Performance Computing Summer School; Athens, Greece</i>	

Cosmology Course Designer and Instructor <i>Spark, MIT Educational Studies Program</i>	Spring 2021
Catalyst Computer Science Instructor <i>Citizen Schools and Mass STEM Hub</i>	Spring 2019
PHYS0220/0270 Astronomy Teaching Assistant <i>Physics Department, Brown University</i>	Fall 2014 - Spring 2015
PHYS0030 Introductory Physics Workshop Assistant <i>Physics Department, Brown University</i>	Fall 2014

LEADERSHIP & SERVICE

Leader & Mentor to Astronomy Graduate Students <i>Astronomy Mentorship Program for Upcoming Postdocs (AMP-UP)</i>	2023 - present
Subject-matter expert reviewer in NASA peer review <i>NASA</i>	2025 - present
Crew Scientist for Analog Astronaut Mission <i>Mars Desert Research Station</i>	Nov - Dec 2025
Pi Day Outreach Organizer <i>Harvard University</i>	2024 - 2025
Reviewer for JOSS <i>Journal of Open Source Software</i>	2022 - present
Graduate Women in Physics Leader <i>Massachusetts Institute of Technology</i>	2019 - 2023
President, MIT Salsa Club <i>Massachusetts Institute of Technology</i>	2018 - 2022
Physics Representative, Diversity and Inclusion Committee <i>MIT Graduate Student Council</i>	2018 - 2022
Treasurer, Students for the Exploration and Development of Space <i>Massachusetts Institute of Technology</i>	2017 - 2021
Adopt-a-Physicist Volunteer <i>Sigma Pi Sigma</i>	2018 - 2020
Observer & Designer, MIT Sidewalk Astrogazers <i>MIT Kavli Institute for Astrophysics and Space Research</i>	2017 - 2019
Head of Design Team, The Triple Helix Magazine <i>Brown University</i>	2014 - 2017
Women in Physics Co-coordinator <i>Brown University</i>	2016 - 2017
Physics Show Presenter <i>Physics & Astronomy Department, Texas A&M University</i>	Summer 2014